Lakshya JEE 2.0 (2024)

Application of Derivative

DPP-02

1. The equation of the tangent to the curve $y = e^{-|x|}$ at the point, where the curve cuts the line $x = 1$ is
(1) $e(x + y) = 1$
(2) $y = ex = 1$
(3) $x + y = e$
(4) None of these

2. The distance between the origin and the normal to the curve $y = e^{2x} + x^2$ at $x = 0$ is
(1) 2
(2) $\frac{2}{\sqrt{3}}$
(3) $\frac{2}{\sqrt{5}}$
(4) $\frac{1}{2}$

3. The number of tangents that can be drawn from (2, 0) to the curve $y = x^6$ is/are
(1) 0
(2) 1
(3) 2
(4) 3

4. The number of tangents to the curve $y^2 - 2x^3 - 4y + 8 = 0$ that pass through (1, 2) is
(1) 3
(2) 1
(3) 2
(4) 6

5. If the curves $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$ and $y^3 = 16x$ intersects at right angles, then $3a^2$ is equal to
(1) 1
(2) 2
(3) 3
(4) 4

6. If curve $y = 1 - ax^2$ and $y = x^2$ intersect orthogonally then a is
(1) $\frac{1}{2}$
(2) $\frac{1}{3}$
(3) 2
(4) 3

7. The area of a triangle formed by a tangent to the curve $2xy = a^2$ and the coordinate axes, is
(1) $2a^2$
(2) a^2
(3) $3a^2$
(4) None of these

8. If the tangent at any point $P(4m^2, 8m^3)$ of $x^3 - y^2 = 0$ is a normal also the curve $x^3 - y^2 = 0$ then find the value of $(9m^2 + 2)$.

9. Find the value of c such that the line joining the points (0, 3) & (5, –2) becomes tangent to the curve $y = \frac{c}{x+1}$.

10. Let C be the curve $y = x^3$ (where x takes all real values). The tangent at A except (0, 0) meets the curve again at B. If the gradient at B is k times the gradient at A, then k is equal to
(1) 4
(2) 2
(3) – 2
(4) $\frac{1}{4}$

11. A curve is represented by the equation. $x = \sec^2 t$ and $y = \cot t$ where t is the parameter, tangent at the point P on the curve where $t = \frac{\pi}{4}$ meets the curve again at the point Q then, $|PQ|$ is equal to:
(1) $\frac{5\sqrt{3}}{2}$
(2) $\frac{5\sqrt{5}}{2}$
(3) $\frac{2\sqrt{5}}{3}$
(4) $\frac{3\sqrt{5}}{2}$

12. The normal of the curve $x = a(\cos\theta + \theta\sin\theta)$ $y = a(\sin\theta - \theta\cos\theta)$ at any θ is such that
(1) It makes a constant angle with x-axis
(2) It passes through the origin
(3) It is at a constant distance from the origin
(4) None of these

13. If the curves $y=a^x$ and $y=e^x$ intersect at an angle α, then tan α equals:

(1) $\left|\frac{\log_e a}{1+\log_e a}\right|$ (2) $\left|\frac{1+\log_e a}{1-\log_e a}\right|$

(3) $\left|\frac{\log_e a-1}{\log_e a+1}\right|$ (4) none of these

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (4)
2. (3)
3. (3)
4. (3)
5. (4)
6. (2)
7. (2)
8. (4)
9. (4)
10. (1)
11. (4)
12. (3)
13. (3)